# Ashwin Balaji Arun Kumar

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**Location:** Dortmund, Germany

## Education

**Fachhochschule Dortmund – University of Applied Sciences and Arts** Dortmund, DE
*Master of Engineering (M.Eng.) in Embedded Systems Engineering* 10/2025 – Current
*(Results pending for first semester)*

**Anna University** Coimbatore, India
*Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering* 08/2018 – 06/2022
*GPA: 1.5 / 4.0 (German equivalent)*

## Skills

**Languages & Tools:** C, C++, Python, Assembly (Fundamentals), Matlab/Simulink, Git, SVN
**Hardware:** TI C2000 (TMS320F), Renesas RL78, STM32 (F4 Series), ESP32, RP2040
**Communication Protocols:** I2C, UART, SPI, Modbus, RS485, MQTT
**Build Systems:** Latex, Linker Scripts, Make, CMake, Makefiles
**Quality & Standards:** Static Analysis (Coverity), Failure Mode and Effects Analysis (FMEA), IEC standards

## Work Experience

**Schneider Electric** Mumbai, India
*Deputy Manager – Firmware* 08/2023 – 09/2025

- Spearheaded **3-20 HP VFD** firmware development in C, managing the full lifecycle on TI and RL78 MCUs.
- **Architected** a modular application layer, decoupling control logic from hardware using **HAL** for portability.
- Developed a high-efficiency (98%) **P&O MPPT** algorithm for DC solar pumping systems.
- Optimized **short-circuit protection** to **200ns** using Trip Zones, thereby **reducing field failures.**
- Integrated **static analysis** into a CI/CD pipeline, ensuring **10+** high-quality firmware releases.
- Implemented flash-based EEPROM emulation to replace EEPROM, significantly reducing unit **BOM cost**.

*Graduate Engineer Trainee – Electromechanical* 07/2022 – 08/2023

- Supported **6A – 100A RCBO** range lifecycle by managing Design changes and customer complaints.
- Improved Isolators thermal performance (IEC 60947-3) by achieving a **7°C reduction in temperature rise**.
- Identified 30+ failure cases in DFMEA and **resolved 2 failures** with RPN>144.

## Projects

**i-Practace – Automatic Tennis ball throwing machine**

- Secured funding of **€9,900** (Rs.9,48 lakhs) grant from Department of Science & Technology, Govt of India
- Managed the overall POC Development, including hardware selection, AC/DC motor control, sensor integration, firmware and IoT development.

## Patents and Publications

- **Automated Tennis game coaching device** - Indian Design Patent Number - 367774-001 (Journal No-20/2023)
- Comparative Analysis on Industrial IoT Communication Protocols and its Future Directives - IEEE XPLORE

## Languages

**German:** A2 (Actively pursuing B1) **English:** C1